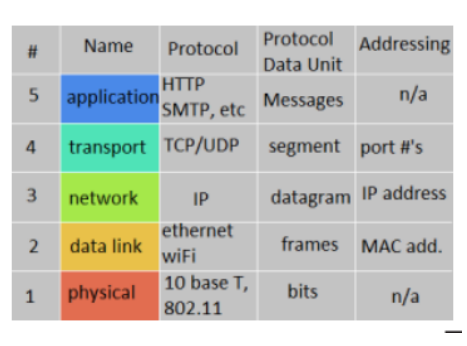
# EE6403 Distributed Multimedia Systems

## Part 2.1: Media Transmission and Delivery Introduction

- TCP/IP 5 Layer Model



| # | Name | Protocol | Protocol Data Unit | Addressing |
|---|---|---|---|---|
| 5 | application | HTTP SMTP, etc | Messages | n/a |
| 4 | transport | TCP/UDP | segment | port #'s |
| 3 | network | IP | datagram | IP address |
| 2 | data link | ethernet WiFi | frames | MAC add. |
| 1 | physical | 10 base T, 802.11 | bits | n/a |

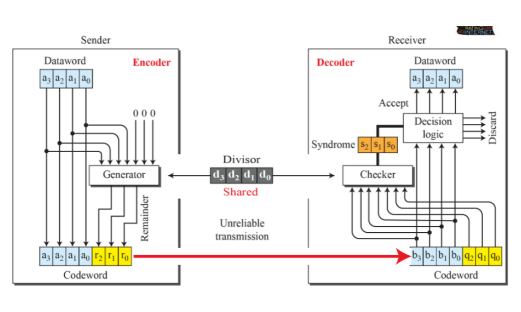
- four sources of packets delay
  - $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$
  - $d_{trans} = L/R$
  - $d_{prop} = d/s$

## Part 2.2: Physical Layer & Media

- DATA AND SIGNAL
  - Analog and Digital
  - Transmission Impairment — $Loss\ (\mathrm{dB}) = 10\log_{10}\frac{P_{in}}{P_{out}}$
  - Data Rate Limits
    - Noiseless Channel: Nyquist Bit Rate — $R = 2B\log_2 M$
    - Noisy Channel: Shannon Capacity — $C = B\log_2(1+SNR)$ — 此SNR单位不是dB
- DIGITAL TRANSMISSION 22
- ANALOG TRANSMISSION 6
- BANDWIDTH UTILIZATION 13
- TRANSMISSION MEDIA 8

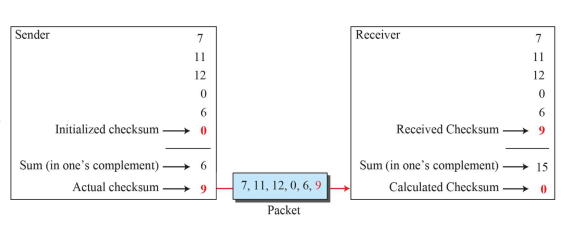
## Part 2.3: Data Link Layer & Technologies

- INTRODUCTION 7
- DATA LINK CONTROL (DLC)
  - Framing — pack bits into frames
  - Flow and Error Control
  - Error Detection and Correction
    - Simple parity-check code
    - Hamming distance
      - codeword with s errors can be detected when s < dmin
      - minimum hamming distance
        - linear code：找非零 codeword 中最少的1个数
        - 普通 code：两两比较，取最小
    - CRC code — sender中补0个数 = divisor最高位级数



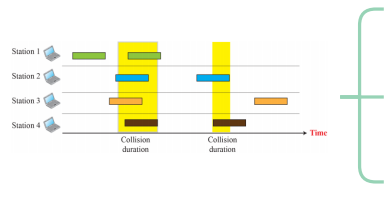
Sender
Encoder
Dataword
Generator
Remainder
Codeword
Divisor
Shared
Unreliable transmission
Receiver
Decoder
Dataword
Accept
Decision logic
Discard
Syndrome
Checker
Codeword

    - Checksum

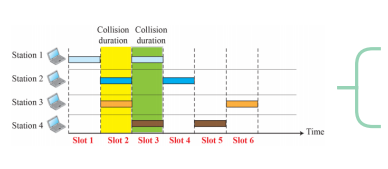


Sender
7
11
12
0
6
Initialized checksum 0
Sum (in one's complement) 6
Actual checksum 9
7, 11, 12, 0, 6, 9
Packet
Receiver
7
11
12
0
6
Received Checksum 9
Sum (in one's complement) 15
Calculated Checksum 0

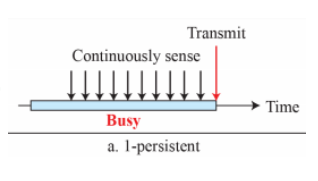
- MULTIPLE ACCESS PROTOCOLS (MAC)
  - random access
    - ALOHA
      - Pure ALOHA



        - $T_B$: (Back-off time): $R \times T_p$ or $R \times T_{fr}$
          $R$: (Random number): 0 to $2^K - 1$
        - Vulnerable time $= 2 \times T_{fr}$
        - offered load: G = λT
          throughput: S = G × e^(-2G) — λ = frame generation rate，单位 frames/s
          T = frame transmission time，单位 s
      - Slotted ALOHA

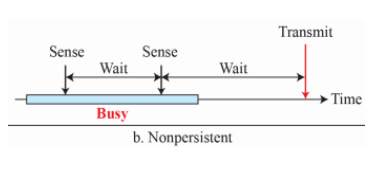


        - Vulnerable time $= T_{fr}$
        - offered load: G = λT
          throughput: S = G × e^(-G)
    - CSMA
      - vulnerable time = propagation time
      - three persistence methods



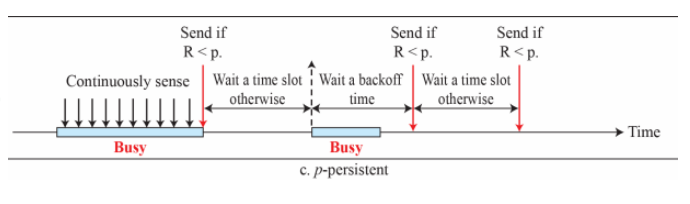
Transmit
Continuously sense
Time
Busy

a. 1-persistent



Sense
Wait
Sense
Wait
Transmit
Time
Busy

b. Nonpersistent



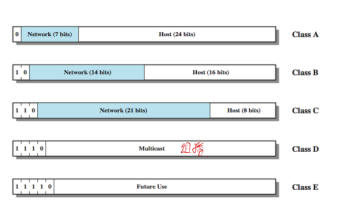
Continuously sense
Send if R < p.
Wait a time slot otherwise
Send if R < p.
Wait a backoff time
Send if R < p.
Wait a time slot otherwise
Time
Busy
Busy

c. p-persistent

      - CSMA/CD — $T_{fr} = 2 \times T_p$
  - control access
    - polling
    - reservation
    - token passing
  - channelization
    - FDMA
    - TDMA
    - CDMA

## Part 2.4: Network Layer

- IPv4 addressing
  - IP address: 32-bit
  - IP addresses associated with each interface
  - Classes of IP Addresses



Class A
Class B
Class C
Class D
Class E
Multicast
组播

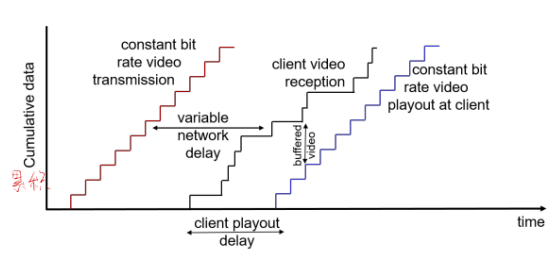
  - network mask
  - Subnets — can physically reach each other without intervening router
  - CIDR: Classless InterDomain Routing — subnet part / host part: 11001000 00010111 00010000 00000000
    200.23.16.0/23

## Part 2.5: Transport Layer

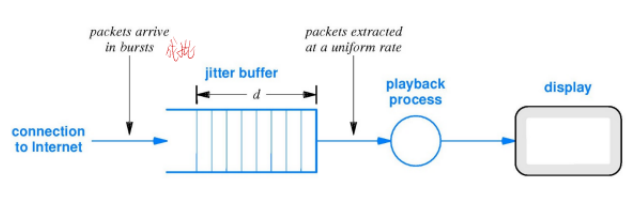
- TCP vs UDP
  - User Datagram Protocol
    - Smaller packet sizes — 8bytes
    - Connectionless
    - Fewer TCP-like controls
    - Primitive error detection
    - No compensation for lost packets
    - Packets arrives out of order — 16-bit checksum
    - No congestion control
    - message-oriented
    - application
      - DNS lookups
      - bandwidth-intensve app
      - video conference
  - Transmission Control Protocol
    - Three-way handshake
    - Retransmission
    - In order delivery
    - Congestion control
    - Bigger headers
    - Introduce latency when network is congested
    - Bigger overhead (retransmission/ACK/jazz)
    - stream-oriented
    - application
      - Text communication
      - file transfer
      - remote access (SSH)
      - firewall
      - multimedia streaming with sufficient bandwidth

## Part 2.6: Quality of Service

- QoS
  - Constant Bit Rate (CBR)
  - streaming stored video

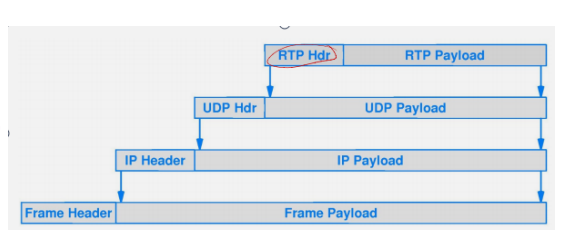


  - Jitter buffer



packets arrive in bursts
缓冲
packets extracted at a uniform rate
jitter buffer
connection to Internet
playback process
display

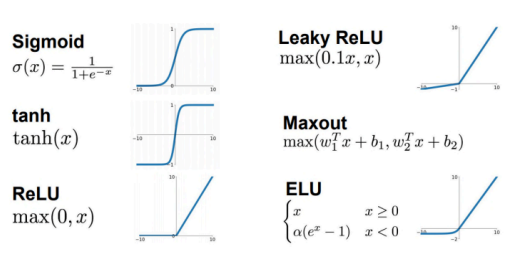
  - Real-Time Transport Protocol (RTP)



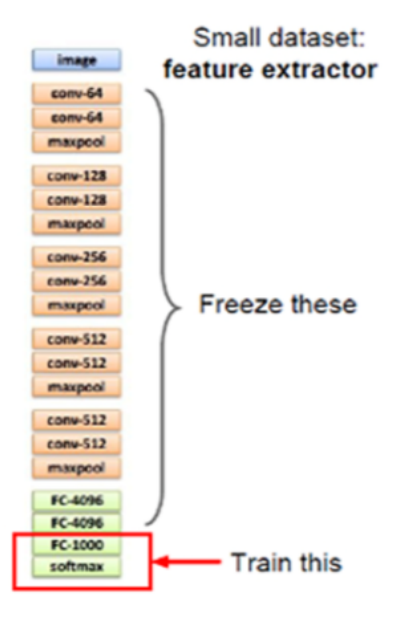
RTP Hdr
RTP Payload
UDP Hdr
UDP Payload
IP Header
IP Payload
Frame Header
Frame Payload

## Part1.4: Multimedia Application

- Linear Classifiers & Artificial Neural Networks (ANNs)
  - $f(x) = Wx + b$
  - common regression loss functions
    - Square loss: $L(x,y) = \sum_i (y_i - f(x_i))^2$
    - Mean Square Error (MSE): $MSE = \frac{1}{N}\sum_i (y_i - f(x_i))^2$
    - Mean Absolute Error (MAE): $MAE = \frac{1}{N}\sum_i |y_i - f(x_i)|$
  - softmax loss (common classification loss function)
    - $p_j = \frac{e^{z_j}}{\sum_k e^{z_k}}$, where $z_j = f(x_j)$
    - $L = -\sum_j y_j \log_e p_j$
- Convolutional Neural Network (CNN)
  - volume size of Conv layer: $W_2 = (W_1 - F + 2P)/S + 1$
  - activation layer

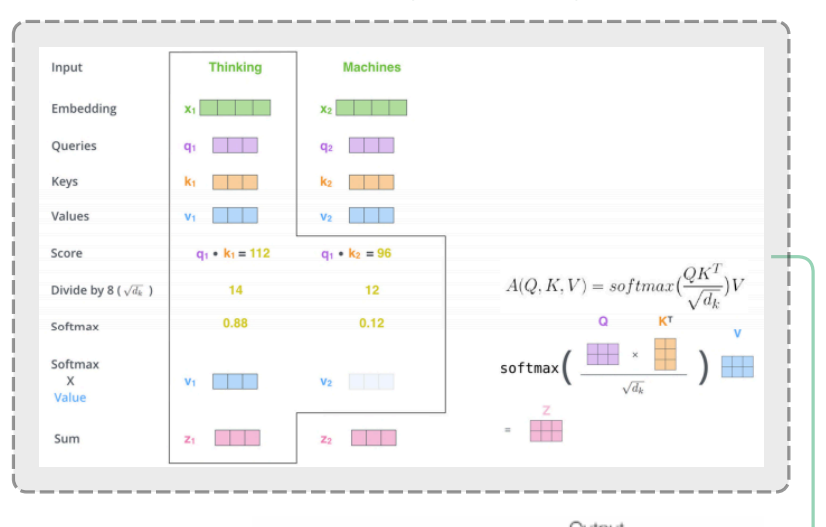


- Transfer Learning

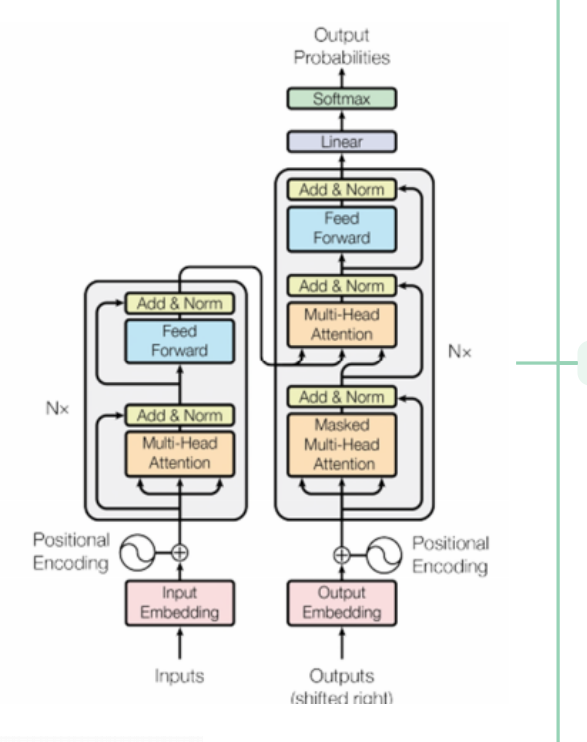


Small dataset:
feature extractor
image
conv-64
conv-64
maxpool
conv-128
conv-128
maxpool
conv-256
conv-256
maxpool
conv-512
conv-512
maxpool
conv-512
conv-512
maxpool
FC-4096
FC-4096
FC-1000
softmax
Freeze these
Train this

  (1) Choose a pretrained network trained on available large dataset
  (2) Modify the last layer from 1000 output nodes to 5 output nodes
  (3) Freeze the early layers and train only the late layers of the pretrained model.
  (4) Apply the trained network after transfer learning to the test images
- Transformer

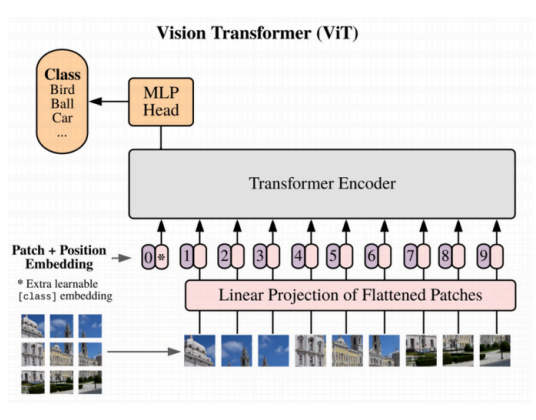


Input
Thinking
Machines
Embedding
Queries
Keys
Values
Score
q1 · k1 = 112
q1 · k2 = 96
Divide by 8 (√d_k)
14
12
Softmax
0.88
0.12
Softmax X Value
Sum
A(Q, K, V) = softmax(QK^T/√d_k)V
softmax



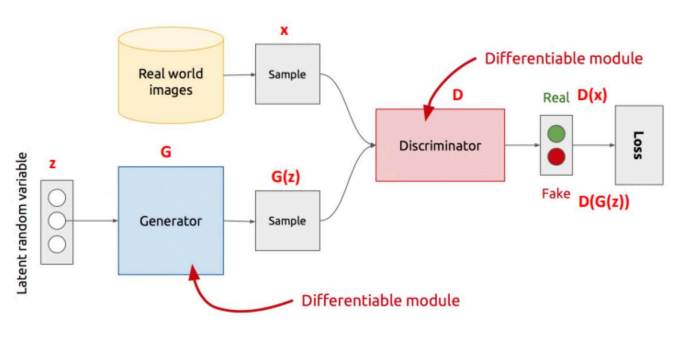
Output Probabilities
Softmax
Linear
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Add & Norm
Masked Multi-Head Attention
Nx
Nx
Positional Encoding
Positional Encoding
Input Embedding
Output Embedding
Inputs
Outputs (shifted right)

  - Vision Transformer (ViT)



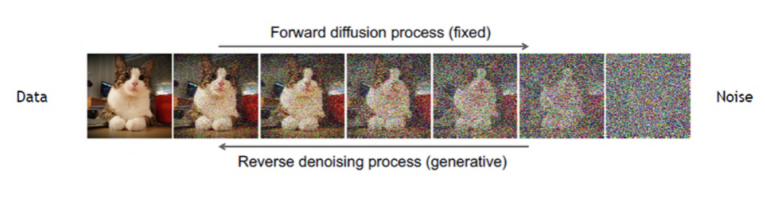
Vision Transformer (ViT)
Class
Bird
Ball
Car
MLP Head
Transformer Encoder
Patch + Position Embedding
* Extra learnable [class] embedding
Linear Projection of Flattened Patches

- Generative AI
  - Generative Adversarial Network (GAN)



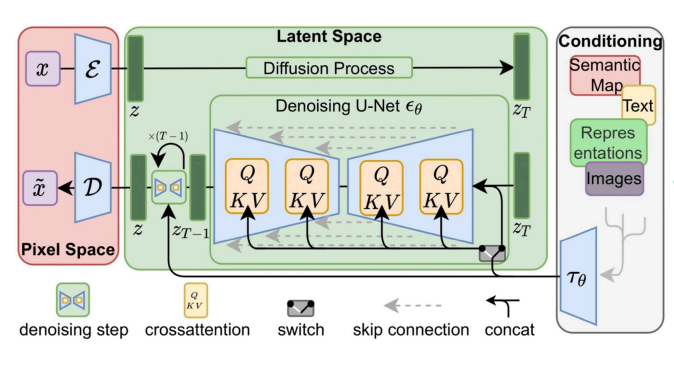
Real world images
x
Sample
Differentiable module
D
Discriminator
Real
D(x)
Fake
D(G(z))
Loss
Latent random variable
z
G
Generator
G(z)
Sample
Differentiable module

  - Diffusion Model



Forward diffusion process (fixed)
Data
Noise
Reverse denoising process (generative)

    - Denoising Diffusion Probabilistic Models (DDPM)
    - Text-Conditioned Image Generation (Text-to-Image)
  - Stable Diffusion

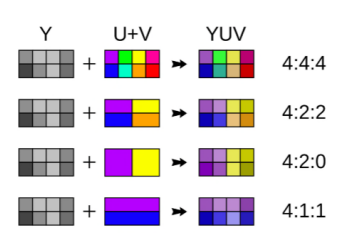


Latent Space
Diffusion Process
Denoising U-Net ε_θ
Pixel Space
Conditioning
Semantic Map
Text
Representations
Images
denoising step
crossattention
switch
skip connection
concat

    - VAE encoder
    - U-Net
    - VAE decoder

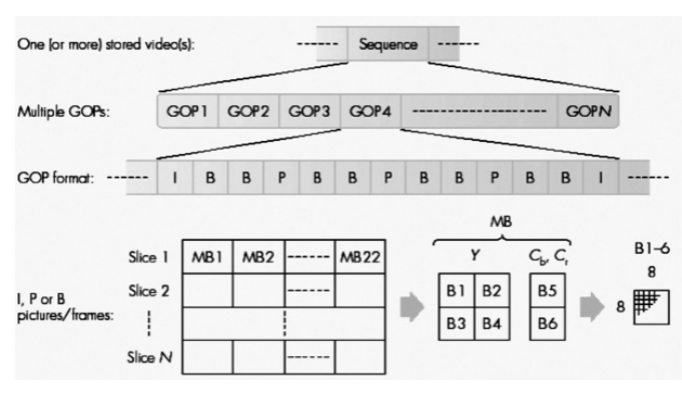
## Part1.3: Video Compression & Standards, Digital Audio, Media Storage & Display

- Video Coding & Standards
  - intraframe compression — JPEG-based compression
  - interframe compression
    - Chroma Subsampling



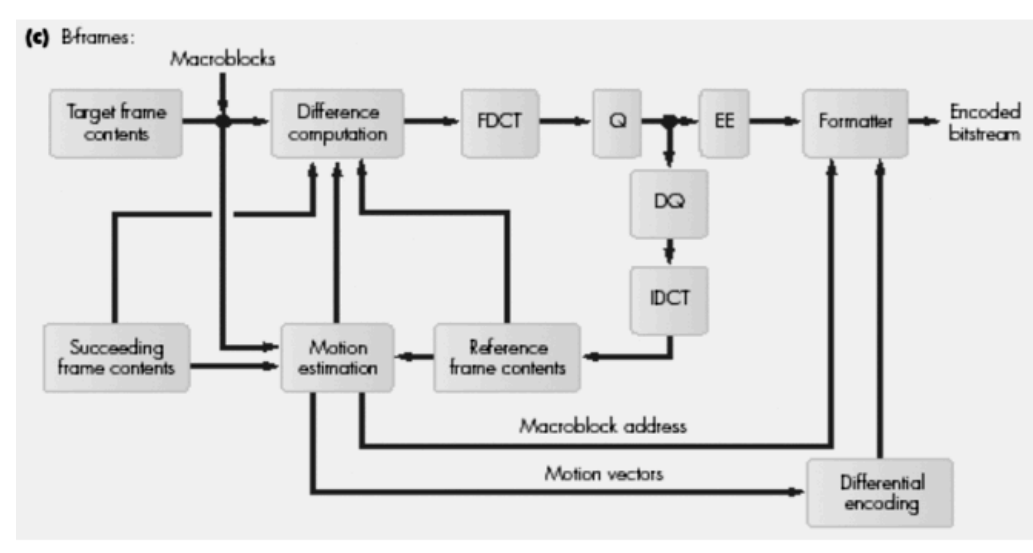
Y
U+V
YUV
4:4:4
4:2:2
4:2:0
4:1:1

    - motion estimation
      - motion vector — reference macroblock -> matched macroblock
      - $MAD(i,j) = \frac{1}{N^2}\sum_{k=0}^{N-1}\sum_{l=0}^{N-1}|C(x+k,y+l) - R(x+i+k,y+j+l)|$
    - motion compensation — prediction error — Mtarget-Mreference
- I, P & B Frame Encoding



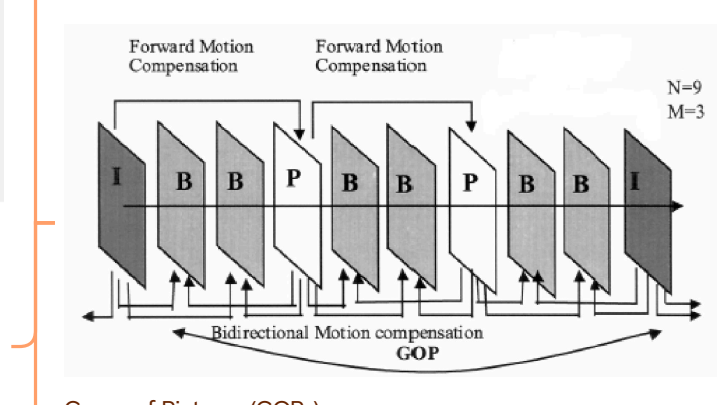
One (or more) stored video(s):
Sequence
Multiple GOPs:
GOP1
GOP2
GOP3
GOP4
GOPN
GOP format:
I B B P B B P B B P B B I
Slice 1
Slice 2
Slice N
MB1
MB2
MB22
I, P or B pictures/frames:
MB
Y
Cb Cr
B1 B2 B3 B4 B5 B6
B1-6
8
8

  MPEG Video structure



(c) B-frames:
Macroblocks
Target frame contents
Difference computation
FDCT
Q
EE
Formatter
Encoded bitstream
DQ
IDCT
Succeeding frame contents
Motion estimation
Reference frame contents
Macroblock address
Motion vectors
Differential encoding

  B-Frame Encoding Flowchart

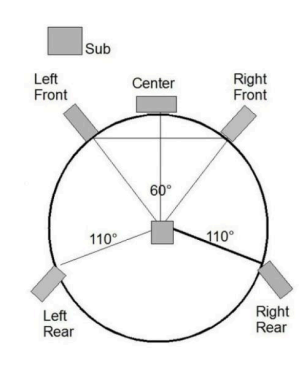


Forward Motion Compensation
Forward Motion Compensation
N=9
M=3
I B B P B B P B B I
Bidirectional Motion compensation
GOP

  Group of Pictures (GOPs)

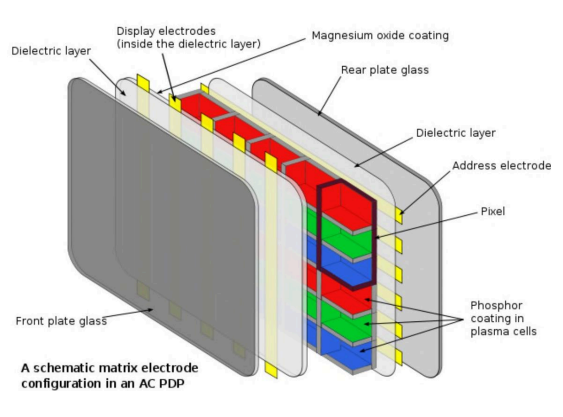
  uncompressed frame data = width * height * bits/pixel * (1+2a)
  a = 0.25, for 4:2:0 or 4:1:1
  a = 0.5, for 4:2:2
  a = 1, for 4:4:4

  the average compression ratio = 1 / the average compression factor
  the average compression factor = I-frame portion / I-frame compression ratio + P-frame portion / P-frame compression ratio + B-frame portion / B-frame compression ratio
- Digital Audio
  - Nyquist Sampling Theorem — Nyquist rate = $2f_{max}$
  - Aliasing
- Dolby Surround Sound



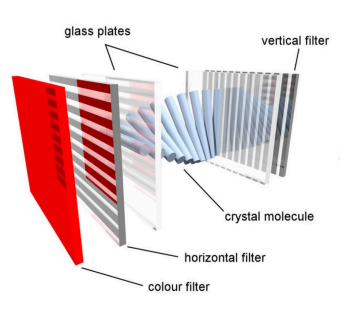
Sub
Left Front
Center
Right Front
60°
110°
110°
Left Rear
Right Rear

  Dolby Digital: 5.1
- Media Storage
  - CDROM
  - Video CD (VCD)
  - DVD
  - Blu-ray Disc
  - Flash Memory
  - Media Streaming
- TV Display
  - Plasma TV



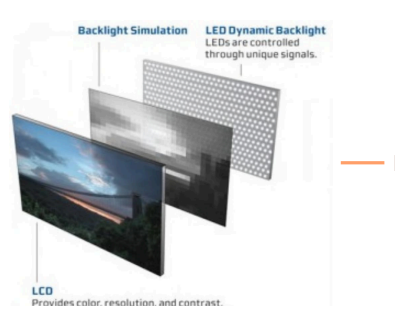
Dielectric layer
Display electrodes (inside the dielectric layer)
Magnesium oxide coating
Rear plate glass
Dielectric layer
Address electrode
Pixel
Phosphor coating in plasma cells
Front plate glass
A schematic matrix electrode configuration in an AC PDP

  - Liquid Crystal Display (LCD) TV



glass plates
vertical filter
crystal molecule
horizontal filter
colour filter

  - LED-Backlight TV

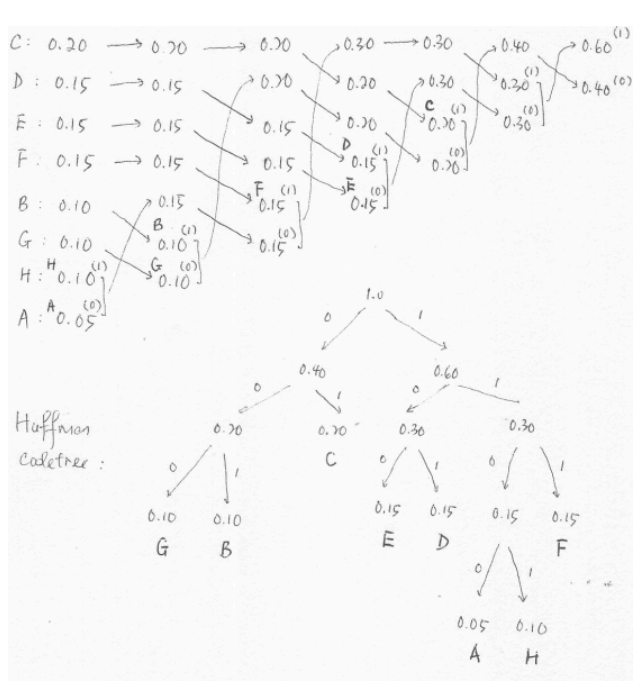


Backlight Simulation
LED Dynamic Backlight
LEDs are controlled through unique signals.
LCD
Provides color, resolution, and contrast

  - Organic Light-Emitting Diode (OLED) TV
  - 3D TV
    - Passive glasses
      - Anaglygh glasses
      - Polarized glasses
    - Active shutter glasses
    - Auto-stereoscopy technology
      - parallax barrier
      - lenticular lens

## Part1.2 Text and Image compression & Standards

- Text compression
  - Huffman coding (Variable Length Coding)



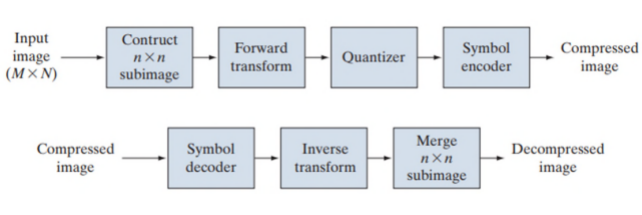
C: 0.20
D: 0.15
E: 0.15
F: 0.15
B: 0.10
G: 0.10
H: 0.10
A: 0.05
1.0
0.40
0.60
0.20
0.20
0.30
0.30
0.10
0.10
0.15
0.15
0.15
0.15
0.05
0.10
G
B
C
E
D
F
A
H
Huffman Codetree

    数值相等时，两个概率的和放在单独概率下
    Tree 左0右1

    A: 1100
    B: 001
    C: 01
    D: 101
    E: 100
    F: 111
    G: 000
    H: 1101

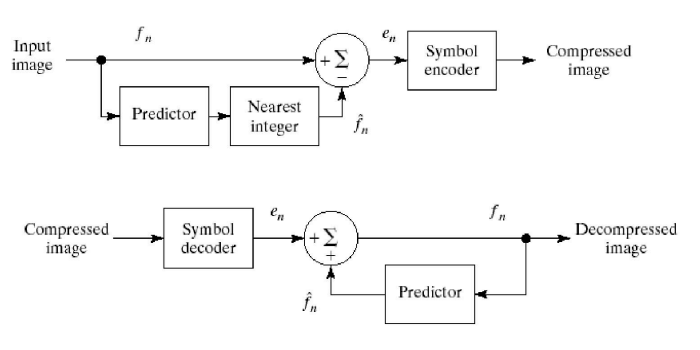
    codeword 由 root node 到leaf node

    compression ratio = uncompressed data length / compressed data length
    >1 effective, <1 ineffective
- Image & Video Compression Basics
  - source coding — compression
    - Statistical Redundancy
      - Spatial redundancy — intraframe redundancy
      - Temporal redundancy — interframe redundancy
      - Coding redundancy
    - Psycho-visual Redundancy
      - Frequency masking
      - Color masking
  - channel coding — error control coding
- Transform-based Coding/Compression



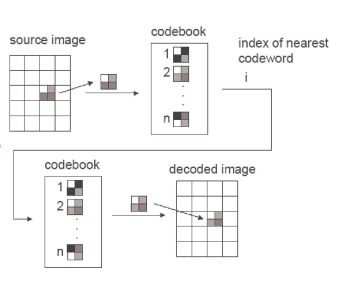
Input image (M × N)
Construct n × n subimage
Forward transform
Quantizer
Symbol encoder
Compressed image
Compressed image
Symbol decoder
Inverse transform
Merge n × n subimage
Decompressed image

- Discrete Cosine Transform (DCT)
  - $S_{uv} = \alpha(u)\alpha(v)\sum_{i=0}^{N-1}\sum_{j=0}^{N-1} s_{ij}\cos\frac{(2i+1)u\pi}{2N}\cos\frac{(2j+1)v\pi}{2N}\quad u,v = 0,\dots,N-1$
  - $\alpha(k) = \begin{cases}\sqrt{\frac{1}{N}} & \text{for } k = 0\\ \sqrt{\frac{2}{N}} & \text{for } k = 1,2,\dots,N-1\end{cases}$
  - DCT原因/优缺点
    - Energy compaction ; Redundancy reduction
    - basis function is fixed and not image-dependent
    - compression is not as effective as KLT
  - $F(u,v) = \mathbf{T}\cdot f(i,j)\cdot\mathbf{T}^T$. We will name $\mathbf{T}$ the *DCT-matrix*.
    $\mathbf{T}[i,j] = \begin{cases}\frac{1}{\sqrt{N}}, & \text{if } i = 0\\ \sqrt{\frac{2}{N}}\cdot\cos\frac{(2j+1)\cdot i\pi}{2N}, & \text{if } i > 0\end{cases}$
    — The 2D IDCT matrix implementation is simply: $f(i,j) = \mathbf{T}^T\cdot F(u,v)\cdot\mathbf{T}$.
- Karhunen-Loeve Transform (KLT)
  - principal component analysis (PCA), eigenvalue decomposition (EVD)
  - Pros and Cons
    - Optimal in terms of energy compaction, coefficient decorrelation and dimensionality reduction.
    - image-dependent
    - requires estimation of image covariance matrix
    - computationally intensive
  - KLT Procedure
    - Compute the covariance matrix:
      $C = E[xx^T] = E[(x_n - \bar{x}_n)(x_n - \bar{x}_n)^T]$
      In practice, this is estimated using sample covariance matrix:
      $C = \frac{1}{N_T - 1}\sum_{i=1}^{N_T}(x_{ni} - \bar{x}_n)(x_{ni} - \bar{x}_n)^T$
      $\bar{x}_n = \frac{1}{N_T}\sum_{i=1}^{N_T} x_{ni}$
    - eigen-decomposition: $Cp_i = \lambda_i p_i$ — $\det(\mathbf{R} - \lambda\mathbf{I}) = 0$
    - Low-Rank Modeling: $\hat{x} = \sum_{i=1}^{r} c_i p_i$
    - Square error: $= \|b - \hat{b}\|^2$
- Differential/Predictive Coding



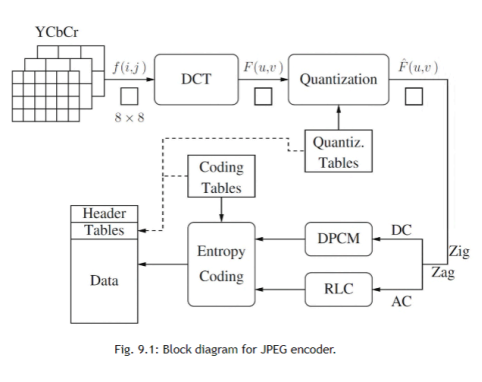
Input image
f_n
e_n
Σ
Symbol encoder
Compressed image
Predictor
Nearest integer
f̂_n
Compressed image
Symbol decoder
e_n
Σ
f_n
Decompressed image
f̂_n
Predictor

- Vector Quantization (VQ)
  - bits per uncompressed symbol = pixel number * bits/pixel



source image
codebook
index of nearest codeword
codebook
decoded image

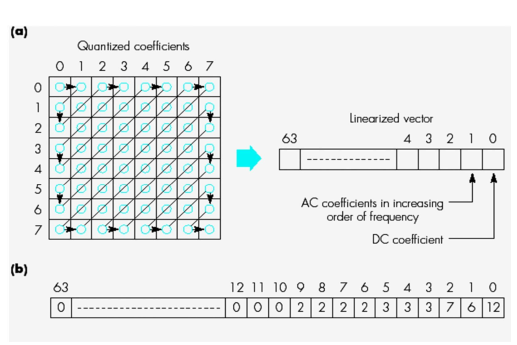
- Singular Value Decomposition (SVD)
  - $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
  - $\mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$
  - $\sigma_i = \sqrt{\lambda_i}$
  - $\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i$
  - $\mathbf{A}^T\mathbf{u}_i = \sigma_i\mathbf{v}_i$
  - $\mathbf{A}\mathbf{A}^T\mathbf{u}_i = \lambda_i\mathbf{u}_i$
  - $\mathbf{A}^T\mathbf{A}\mathbf{v}_i = \lambda_i\mathbf{v}_i$
  - $\hat{\mathbf{A}} = \sum_{i=1}^{p}\sigma_i\mathbf{u}_i\mathbf{v}_i^T$
- JPEG



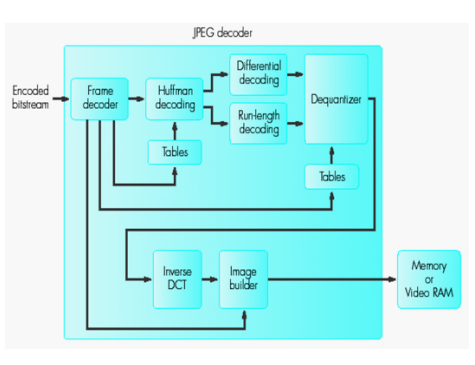
YCbCr
f(i, j)
8 × 8
DCT
F(u, v)
Quantization
F̂(u, v)
Quantiz. Tables
Coding Tables
Header Tables
Data
Entropy Coding
DPCM
RLC
DC
AC
Zig Zag

  Fig. 9.1: Block diagram for JPEG encoder.

  - DPCM: Differential Pulse Code Modulation — 连续图像块 DC 系数差异小的特点
  - RLC: Run Length Coding — 用于对经过之字形扫描后存在大量连续零值的 AC 系数进行编码
  - FDCT: $S_{uv} = \frac{1}{4}C_uC_v\sum_{i=0}^{7}\sum_{j=0}^{7}s_{ij}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}$
  - IDCT: $s_{ij} = \frac{1}{4}\sum_{u=0}^{7}\sum_{v=0}^{7}C_uC_vS_{uv}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}$
  - $C_u, C_v = \begin{cases}\frac{1}{\sqrt{2}} & \text{for } u, v = 0\\ 1 & \text{otherwise}\end{cases}$



(a)
Quantized coefficients
0 1 2 3 4 5 6 7
Linearized vector
63
4 3 2 1 0
AC coefficients in increasing order of frequency
DC coefficient
(b)
63
12 11 10 9 8 7 6 5 4 3 2 1 0



JPEG decoder
Encoded bitstream
Frame decoder
Huffman decoding
Differential decoding
Run-length decoding
Tables
Dequantizer
Tables
Inverse DCT
Image builder
Memory or Video RAM

  - JPEG Modes
    - Sequential DCT Mode
    - Progressive DCT Mode
    - Hierarchical Mode
    - Lossless Mode